

AdventureWorks Sales & Profit Analysis

Power BI Portfolio Project Documentation

Dashboard KPI	Displayed Value
Total Sales	\$110M
Gross Profit	\$13M
Gross Margin	11.43%
Total Orders	31K

This report documents an interactive Power BI dashboard built with the AdventureWorks dataset. The project demonstrates data preparation, data modeling, DAX measures, time-intelligence analysis, interactive filtering, and business-focused visualization.

1. Project Overview

The objective is to provide an executive-friendly view of sales performance, profitability, product performance, customer contribution, geography, and sales trends. The report uses Power BI visuals and DAX measures to turn transactional data into decision-support information.

2. Business Questions

- What are total sales, gross profit, gross margin, and order volume?
- How do current-year sales compare with the previous year?
- Which sales channels and regions contribute most to revenue?
- Which product categories and products drive sales and profit?
- Which customers and countries contribute most to sales?
- How do sales, quantity, and average order value vary by category and subcategory?

3. Tools & Technologies

Microsoft Power BI for dashboarding and visualization; **Power Query** for data preparation and transformation; **DAX** for calculated measures and time intelligence; and **data modeling** for connecting analytical tables.

4. Dashboard Structure

Executive Overview: KPI cards, sales trend, sales channel, regional sales, gross profit by category, and top products.

Product & Customer Analysis: category performance, top customers, country sales, quantity sold, average order value, and subcategory sales.

DAX Validation: a validation table showing fiscal-year results for the major measures, supporting measure verification.

5. Executive Overview Dashboard

The executive page summarizes the business with four headline KPIs. The displayed dashboard shows approximately \$110M in total sales, \$13M in gross profit, an 11.43% gross margin, and 31K total orders. The page also compares sales over time and breaks performance down by channel, region, product category, and top products.

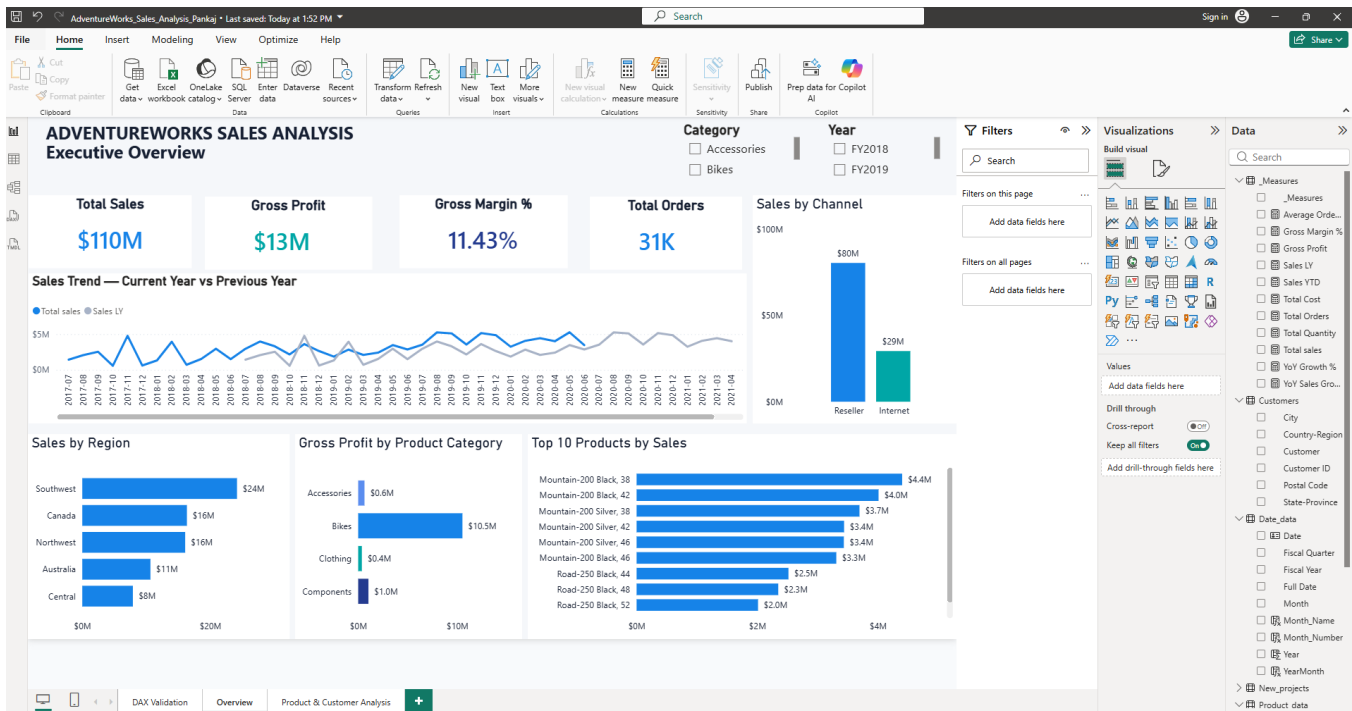


Figure 1. AdventureWorks Executive Overview dashboard.

6. Product & Customer Analysis

The second page focuses on product and customer contribution. The dashboard shows Bikes as the dominant product category at about \$95M in sales, followed by Components at about \$12M. It also provides rankings for top customers, countries, subcategories, quantity sold, and average order value, with Year and Category slicers for interactive analysis.

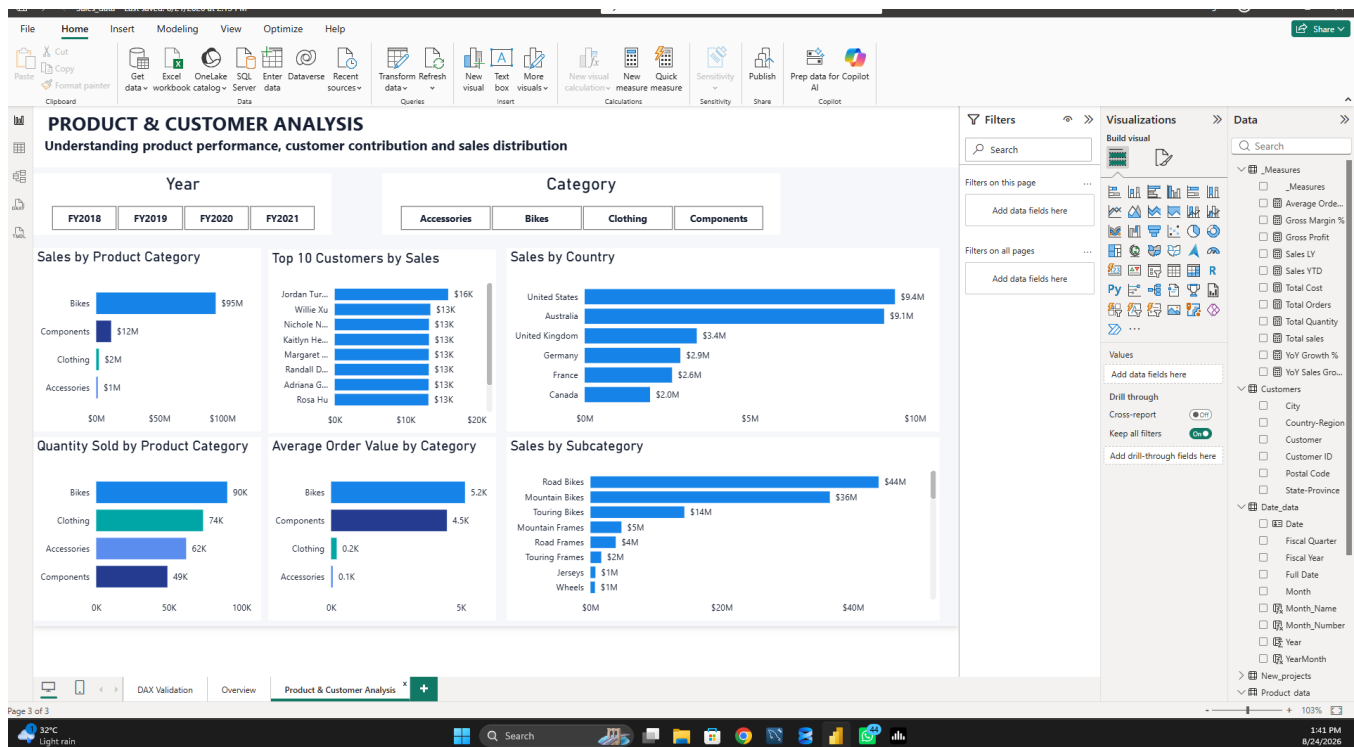


Figure 2. Product & Customer Analysis dashboard.

7. DAX Measures & Validation

The project includes measures for Total Sales, Total Cost, Gross Profit, Gross Margin %, Total Quantity, Total Orders, Average Orders Value, Sales YTD, Sales LY, YoY Sales Growth, and YoY Growth %. The validation page places these measures together by Fiscal Year so that results can be checked under a controlled filter context.

Example calculation: Gross Profit = Total Sales – Total Cost. Time-intelligence measures are used to compare current performance with prior periods and calculate year-over-year movement.

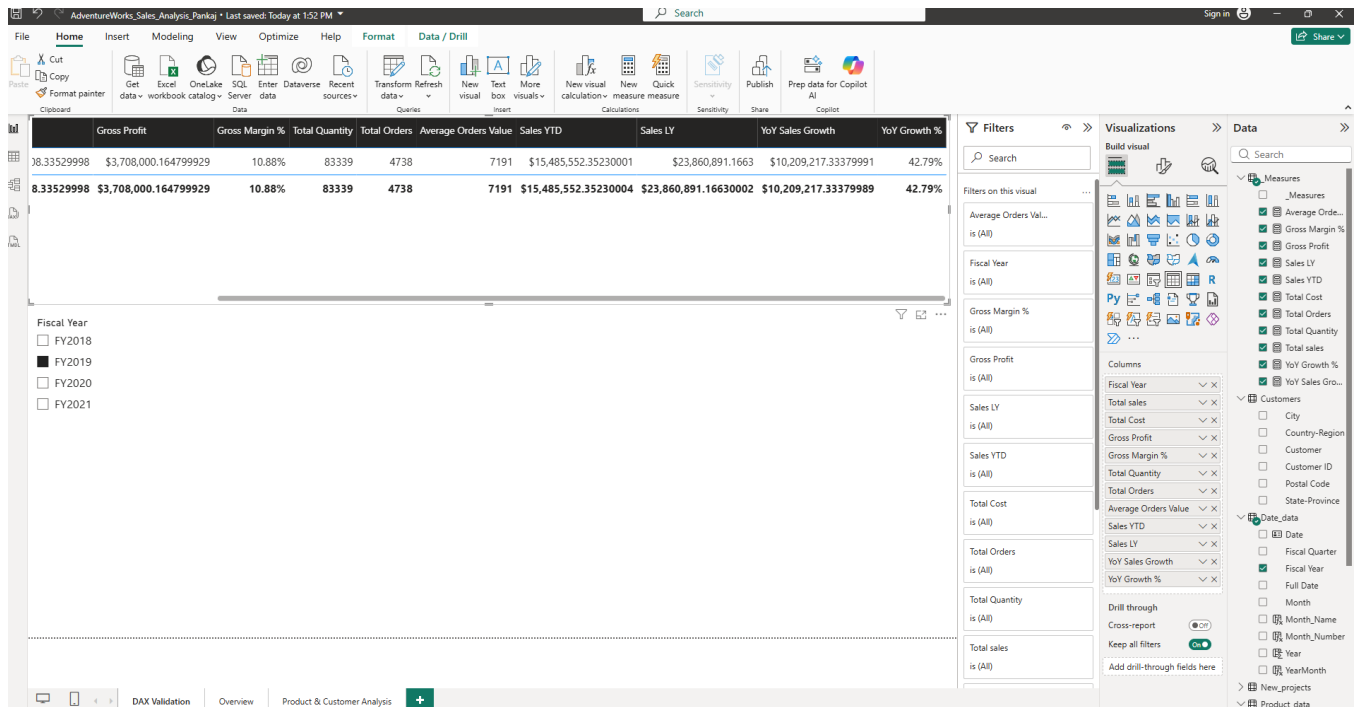


Figure 3. DAX validation page showing measure outputs by fiscal year.

8. Key Business Insights

- **Revenue concentration:** Bikes are the largest product category, contributing roughly \$95M in displayed sales.
- **Channel mix:** Reseller sales are substantially larger than Internet sales on the executive overview, at roughly \$80M versus \$29M.
- **Regional performance:** Southwest is the leading displayed region at about \$24M, with Canada and Northwest around \$16M each.
- **Profitability:** The executive overview reports approximately \$13M gross profit and an 11.43% gross margin.
- **Product contribution:** The top-products visual is dominated by Mountain-200 and Road-250 products, indicating strong concentration among leading products.
- **Interactive analysis:** Fiscal year and category selectors allow users to move from a high-level executive view to focused product and customer analysis.

9. Skills Demonstrated

- Power BI dashboard development
- Power Query data transformation
- DAX measure creation
- Time-intelligence analysis
- Data modeling and relationships
- KPI design and business reporting
- Interactive slicers and filtering
- Sales, profitability, product, customer, and geographic analysis

10. Conclusion

The AdventureWorks project demonstrates an end-to-end business intelligence workflow: preparing data, modeling it for analysis, creating reusable DAX measures, validating calculations, and presenting results through an interactive dashboard. The final report is designed to help stakeholders quickly understand sales scale, profitability, product performance, customer contribution, and changes over time.